

*Remark 2.7.* As a consequence of the explicit formula of the previous remark, one can also study the regularity of this map  $T_{\text{mon}}$  depending on the regularity of  $\mu$  and  $\nu$ . Indeed, as soon as these measures are fully supported and have no atoms, the two functions  $F_\mu$  and  $F_\nu$  are homeomorphisms, and so is  $T_{\text{mon}}$ . Moreover, if  $\mu$  and  $\nu$  are absolutely continuous, with continuous densities which do not vanish, then they are also diffeomorphisms, and so is  $T_{\text{mon}}$ . In general, the regularity of  $T_{\text{mon}}$  is one degree higher than that of the two measures, as it is the case for  $F_\mu, F_\nu, F_\mu^{-1}$  and  $F_\nu^{-1}$ .

In the next section we will see that the map  $T_{\text{mon}}$  that we have just built optimizes a whole class of transport costs. To prove it, we will need the following characterization of  $\gamma_{\text{mon}}$  and  $T_{\text{mon}}$ .

**Lemma 2.8.** *Let  $\gamma \in \Pi(\mu, \nu)$  be a transport plan between two measures  $\mu, \nu \in \mathcal{P}(\mathbb{R})$ . Suppose that it satisfies the property*

$$(x, y), (x', y') \in \text{spt}(\gamma), x < x' \Rightarrow y \leq y'. \quad (2.2)$$

*Then, we have  $\gamma = \gamma_{\text{mon}}$ . In particular, there is a unique  $\gamma$  satisfying (2.2).*

*Moreover, if  $\mu$  is atomless, then  $\gamma = \gamma_{\Gamma_{\text{mon}}}$ .*

*Proof.* For the first part of the statement we just need to prove

$$\gamma((-\infty, a] \times (-\infty, b]) = F_\mu(a) \wedge F_\nu(b).$$

Indeed, from Proposition 2.2, this condition is satisfied by  $\gamma_{\text{mon}}$ , and we saw that this is enough to characterize a measure on  $\mathbb{R}^2$ .

Consider the two sets  $A = ]-\infty, a] \times ]b, +\infty[$  and  $B = ]a, +\infty[ \times ]-\infty, b]$ . From the assumption on  $\gamma$ , it is not possible to have both  $\gamma(A) > 0$  and  $\gamma(B) > 0$  (otherwise we would have points in  $\text{spt}(\gamma)$  violating the condition (2.2)). Hence, we can write

$$\gamma((-\infty, a] \times (-\infty, b]) = \gamma((( -\infty, a] \times (-\infty, b]) \cup A) \wedge \gamma((( -\infty, a] \times (-\infty, b]) \cup B).$$

Yet, we have

$$\gamma((( -\infty, a] \times (-\infty, b]) \cup A) = \gamma((-\infty, a] \times \mathbb{R}) = F_\mu(a)$$

and

$$\gamma((( -\infty, a] \times (-\infty, b]) \cup B) = \gamma(\mathbb{R} \times (-\infty, b]) = F_\nu(b).$$

This proves the first part of the claim. For the second part, we suppose  $\mu$  to be atomless. For any point  $x \in \mathbb{R}$  one can define the interval  $I_x$  as the minimal interval  $I$  such that  $\text{spt}(\gamma) \cap (\{x\} \times \mathbb{R}) \subset \{x\} \times I$ . This interval can obviously be reduced to a singleton. The assumption on  $\gamma$  implies that the interior of all these intervals are disjoint (and ordered). In particular, there can be at most a countable quantity of points such that  $I_x$  is not a singleton. Since these points are  $\mu$ -negligible (as a consequence of  $\mu$  being atomless), we can define  $\mu$ -a.e. a map  $T$  such that  $\gamma$  is concentrated on the graph of  $T$ . This map will be monotone non-decreasing because of (2.2), and this gives  $T = T_{\text{mon}}$  since we already know the uniqueness of a non-decreasing map with fixed marginals (Theorem 2.5).  $\square$   $\square$

## 2.2 The optimality of the monotone map

Now that we know quite well the properties, definitions and characterizations of the map  $T_{\text{mon}}$  and of the plan  $\gamma_{\text{mon}}$ , we can see that they are, in the 1D case, optimal for several different costs, and not only for the quadratic one. This is really specific to the one-dimensional case, it will not be true in higher dimension. The cost that we will consider will be convex functions of the difference  $x - y$  and, to stress the possible asymmetric behavior of these costs, we prefer to write  $h(y - x)$  instead of  $h(x - y)$ .

**Theorem 2.9.** *Let  $h : \mathbb{R} \rightarrow \mathbb{R}_+$  be a strictly convex function, and  $\mu, \nu \in \mathcal{P}(\mathbb{R})$  be probability measures. Consider the cost  $c(x, y) = h(y - x)$  and suppose that (KP) has a finite value. Then, (KP) has a unique solution, which is given by  $\gamma_{\text{mon}}$ . In the case where  $\mu$  is atomless, this optimal plan is induced by the map  $T_{\text{mon}}$ .*

*Moreover, if the strict convexity assumption is withdrawn and  $h$  is only convex, then the same  $\gamma_{\text{mon}}$  is actually an optimal transport plan, but no uniqueness is guaranteed anymore.*

*Proof.* We will use the fact that the support of any optimal  $\gamma$  is a  $c$ -CM set  $\Gamma$ . This means in particular that for  $(x, y), (x', y') \in \Gamma$  we have

$$h(y - x) + h(y' - x') \leq h(y' - x) + h(y - x'). \quad (2.3)$$

We only need to show that this implies (in the strictly convex case) a monotone behavior : we will actually deduce from (2.3) that  $x < x'$  implies  $y \leq y'$ , and this will allow to conclude from Lemma 2.8.

To prove  $y \leq y'$  suppose by contradiction  $y > y'$  and denote  $a = y - x$ ,  $b = y' - x'$  and  $\delta = x' - x > 0$ . Condition (2.3) reads  $h(a) + h(b) \leq h(b + \delta) + h(a - \delta)$ . Moreover, the assumption  $y' < y$  implies  $b + \delta < a$ . Hence,  $b + \delta$  and  $a - \delta$  are located in the segment between  $b$  and  $a$  (and  $b < a$ ). More precisely, we have

$$b + \delta = (1 - t)b + ta, \quad a - \delta = tb + (1 - t)a, \quad \text{for } t = \frac{\delta}{a - b} \in ]0, 1[.$$

Thus, strict convexity yields

$$\begin{aligned} h(a) + h(b) &\leq h(b + \delta) + h(a - \delta) \\ &< (1 - t)h(b) + th(a) + th(b) + (1 - t)h(a) = h(a) + h(b). \end{aligned}$$

This gives a contradiction and proves the statement in the strictly convex case.

The statement when  $h$  is only convex is trivial if  $h$  is constant (since every  $\gamma$  is optimal) and, if not, it is obtained by approximation. Lemma 2.10 below proves that there exist a sequence of strictly convex functions  $h_\varepsilon$  such that  $h \leq h_\varepsilon \leq (1 + \varepsilon)h + \varepsilon$ . Let us take the transport cost  $c_\varepsilon(x, y) := h_\varepsilon(y - x)$ . In this case we know that  $\gamma_{\text{mon}}$  optimizes the cost  $\int c_\varepsilon d\gamma$  and hence

$$\int h(y - x) d\gamma_{\text{mon}} \leq \int h_\varepsilon(y - x) d\gamma_{\text{mon}} \leq \int h_\varepsilon(y - x) d\gamma \leq \varepsilon + (1 + \varepsilon) \int h(y - x) d\gamma$$

for all  $\gamma \in \Pi(\mu, \nu)$ . Passing to the limit as  $\varepsilon \rightarrow 0$ , we get that  $\gamma_{\text{mon}}$  also optimizes the cost  $c$ .  $\square$   $\square$

**Lemma 2.10.** *For every non-constant convex and positive function  $h : \mathbb{R} \rightarrow \mathbb{R}_+$  and every  $\varepsilon > 0$  there exists  $h_\varepsilon : \mathbb{R} \rightarrow \mathbb{R}_+$  strictly convex, such that  $h \leq h_\varepsilon \leq (1 + \varepsilon)h + \varepsilon$ .*

*Proof.* We just need to prove that there is a strictly convex function  $f : \mathbb{R} \rightarrow \mathbb{R}_+$  such that  $f \leq h + 1$ . Then, we will take  $h_\varepsilon := h + \varepsilon f$ . From the fact that  $h$  is convex, it is bounded from below by an affine function, and we have  $h(t) \geq (at + b)_+$  (we take the positive part since we also know  $h \geq 0$ ). It can be checked that

$$f(t) := \frac{1}{2} \sqrt{4 + (at + b)^2} + \frac{1}{2}(at + b)$$

is strictly convex, and satisfies  $f(t) \leq \frac{1}{2}(2 + |at + b| + (at + b)) = 1 + (at + b)_+$ . □  
□

**Remark 2.11.** Positivity of the function  $h$  is not really necessary, as soon as  $\mu$  and  $\nu$  satisfy some integrability conditions so that  $\inf(\text{KP}) > -\infty$ . For instance, if we have  $\int |x| d\mu(x), \int |y| d\nu(y) < +\infty$ , we can add an affine function to  $h$  and make it convex, and the cost of this affine function does not depend on  $\gamma$  and is finite (see Example 2.14 below).

**Remark 2.12.** We stress that a completely analogous proof could be used to prove that optimal maps for strictly concave costs (of the form  $h(x - y)$  for  $h$  strictly concave) are monotone decreasing, instead of monotone increasing. Compare to what described in Section 3.3.2.

**Remark 2.13.** Also, the optimality of  $\gamma_{\text{mon}}$  is true under more general assumptions, i.e. for some costs which are not of the form  $h(x - y)$ , but satisfy the *twist condition*. See **Ex(10)**.

Easy examples where  $c$  is convex but not strictly convex and  $T_{\text{mon}}$  is not the unique optimal transport map may be built as follows.

**Example 2.14** (Linear costs). Suppose that  $c(x, y) = L(x - y)$ , the map  $a : \mathbb{R}^d \rightarrow \mathbb{R}$  being linear. In this case any transport plan  $\gamma$  is optimal, and any transport map as well. This can be easily seen if one writes

$$\int L(x - y) d\gamma = \int L(x) d\gamma - \int L(y) d\gamma = \int L(x) d\mu - \int L(y) d\nu,$$

which shows that the result does not depend on  $\gamma$  but only on its marginals. This general example works for  $\mu, \nu$  compactly supported (so that we do not have any problem of integrability of  $L(x)$  and  $L(y)$ ), and in any dimension. Hence, also in 1D.

**Example 2.15** (Distance costs on the line). Suppose that  $c(x, y) = |x - y|$ , and that  $\mu, \nu \in \mathcal{P}(\mathbb{R})$  are such that  $\text{sup}spt(\mu) < \text{inf}spt(\nu)$ . In this case as well any transport plan  $\gamma$  is optimal, and any transport map as well. This can be seen by noting that for every  $(x, y) \in \text{spt}(\mu) \times \text{spt}(\nu)$  we have  $c(x, y) = y - x$ , which is again a linear cost.

**Example 2.16** (Book-shifting). Consider  $c(x, y) = |x - y|$ ,  $\mu = \frac{1}{2}\mathcal{L}_{[0,2]}^1$  and  $\nu = \frac{1}{2}\mathcal{L}_{[1,3]}^1$ . Then  $T_{\text{mon}}(x) = x + 1$  is the monotone transport plan transporting  $\mu$  onto  $\nu$ . Its cost is

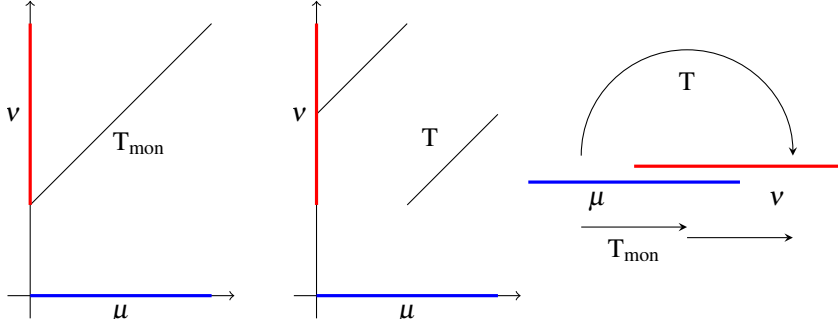


Figure 2.1: The transport maps in the book shifting example

$M(T) = \int |T_{\text{mon}}(x) - x| d\mu = 1$ . Yet, the transport map  $T$  given by

$$T(x) = \begin{cases} x+2 & \text{if } x \leq 1, \\ x & \text{if } x > 1, \end{cases}$$

also satisfies  $T_{\#}\mu = \nu$  and  $\int |T(x) - x| d\mu = \frac{1}{2} \int_0^1 2 dx = 1$ , and is optimal as well.

Starting from the fact that the optimal transport for all these costs is the monotone one we can express the cost for sending a given measure  $\mu$  onto another measure  $\nu$  in terms of their cumulative distribution functions  $F_\mu$  and  $F_\nu$ .

**Proposition 2.17.** *Given  $\mu, \nu \in \mathcal{P}(\mathbb{R})$ , consider the cost  $c(x, y) = h(x - y)$ , where  $h$  is a convex function. Then, we have*

$$\mathcal{T}_c(\mu, \nu) = \int_0^1 h(F_\nu^{[-1]} - F_\mu^{[-1]}) d\mathcal{L}^1.$$

If  $h(z) = |z|$  then this also coincides with  $\int_{\mathbb{R}} |F_\mu(t) - F_\nu(t)| dt$ .

*Proof.* The first part of the statement is just a consequence of the optimality of  $\gamma_{\text{mon}}$  (Theorem 2.9).

The particular case of  $h(z) = |z|$  may be treated by geometric consideration: indeed, the integral  $\int_0^1 |(F_\nu)^{-1} - (F_\mu)^{-1}|$  equals the area of the part of the strip  $[0, 1] \times \mathbb{R}$  bounded by the graphs of  $(F_\mu)^{-1}$  and  $(F_\nu)^{-1}$ . In order to pass from the inverse functions to the direct ones it is enough to turn the head and look at the same strip from the variable  $t$  instead of  $x$ . But, if we want to prove it through computations, we have

$$\begin{aligned} & \int |F_\nu^{[-1]}(x) - F_\mu^{[-1]}(x)| dx \\ &= \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_\nu^{[-1]}(x) \text{ or } F_\nu^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}) \\ &= \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_\nu^{[-1]}(x)\}) \\ &\quad + \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\nu^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}). \end{aligned}$$

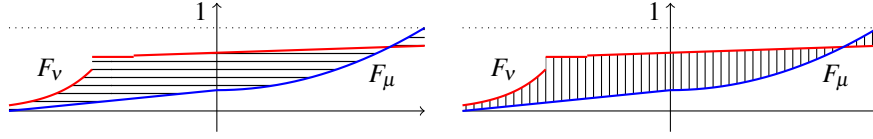


Figure 2.2: The computation of the areas in the last part of Proposition 2.17: the integral of  $|F_V^{[-1]} - F_\mu^{[-1]}|$  corresponds to seeing “horizontally” the graph on the left, while, if seen “vertically”, we get the integral of  $|F_\mu - F_V|$ .

Then, by Fubini’s theorem, we have

$$\begin{aligned} & \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_V^{[-1]}(x)\}) \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : x \leq F_\mu(t) \text{ and } F_V(t) < x\}) dt \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : F_V(t) < x \leq F_\mu(t)\}) dt. \end{aligned}$$

Analogously,

$$\begin{aligned} & \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_V^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}) \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : F_\mu(t) < x \leq F_V(t)\}) dt. \end{aligned}$$

and, summing up,

$$\begin{aligned} & \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_V^{[-1]}(x) \text{ or } F_V^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}) \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : F_V(t) < x \leq F_\mu(t) \text{ or } F_\mu(t) < x \leq F_V(t)\}) dt \\ &= \int_{\mathbb{R}} |F_\mu(t) - F_V(t)| dt. \end{aligned}$$

This concludes the proof.  $\square$

## 2.3 The Knothe transport

The Knothe transport, also known as Knothe-Rosenblatt rearrangement, is a special transport map, which has a priori nothing to do with optimal transport, that may be associated to two given measures  $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ . It was independently proposed by Rosenblatt [264] for statistical purposes and by Knothe [204] for applications to geometric inequalities (see also Section 2.5.3). The main interesting point of such a map is its computational simplicity. We will explain the principle of this transport map